

Detecting Hierarchical Regiment-Sort Structure in Data

Lord Soumadeep Ghosh with ChatGPT (OpenAI)

Kolkata, India

Abstract

This paper develops the inverse problem associated with Regiment Sort: given an observed array, determine whether its structure is consistent with having been generated by a Regiment Sort procedure. The problem combines local monotonicity, group-level aggregation, global ranking, model selection, statistical inference, and robustness to noise and missingness.

For a candidate regiment size, observations are partitioned into equal-sized regiments. Within-regiment ascending order and between-regiment descending aggregate scores are tested separately and then combined into a Regiment Sort Detection Score. When the regiment size is unknown, candidate partitions are compared and the most plausible structure is selected. Permutation and other null-model procedures provide empirical significance tests. Extensions to weighted scores, multidimensional capability, time-series data, financial data, econometrics, machine learning, and hierarchical scientific data are discussed. The resulting framework treats Regiment Sort as a detectable statistical structure rather than merely as a deterministic sorting algorithm.

The paper ends with “The End”

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1 Introduction

Regiment Sort is a hierarchical ordering procedure in which observations are partitioned into equal-sized groups, each group is sorted internally, a group-level aggregate score is calculated, and the groups are subsequently ordered according to that score.

The forward problem is

$$A \longrightarrow \text{RegimentSort}(A). \quad (1)$$

The present paper studies the inverse problem,

$$A_{\text{observed}} \longrightarrow \text{evidence for latent Regiment-Sort structure.} \quad (2)$$

This distinction is fundamental. An arbitrary dataset may display partial monotonicity or group-level clustering without having been generated by Regiment Sort. Consequently, detection should combine structural diagnostics with statistical inference.

The problem is related to grouped data analysis, order statistics, permutation methods, change-point detection, clustering, model selection, and anomaly detection [4, 9, 5].

2 The Inverse Regiment-Sort Problem

Let

$$A = (a_1, \dots, a_N) \quad (3)$$

be an observed array.

Suppose that a candidate regiment size q is proposed, where

$$q = mn \quad (4)$$

and

$$N = rq. \quad (5)$$

The candidate partition is

$$A = R_1 \| R_2 \| \dots \| R_r. \quad (6)$$

For each regiment,

$$R_i = (x_{i1}, \dots, x_{iq}), \quad (7)$$

and its aggregate score is

$$S_i = \sum_{j=1}^q x_{ij}. \quad (8)$$

Exact Regiment Sort structure requires

$$x_{i1} \leq x_{i2} \leq \dots \leq x_{iq} \quad (9)$$

for every i , together with

$$S_1 \geq S_2 \geq \dots \geq S_r. \quad (10)$$

The detection problem is therefore a conjunction of local and global ordering conditions.

3 Exact Detection

Define the local compliance indicator

$$L_i = \mathbf{1} [x_{i1} \leq x_{i2} \leq \dots \leq x_{iq}]. \quad (11)$$

Define global score compliance by

$$G = \mathbf{1} [S_1 \geq S_2 \geq \dots \geq S_r]. \quad (12)$$

An exact detector is

$$D_q^{\text{exact}} = G \prod_{i=1}^r L_i. \quad (13)$$

Thus

$$D_q^{\text{exact}} = 1 \quad (14)$$

if and only if the candidate partition satisfies both defining structural conditions.

4 A Noisy Detection Score

Exact detection is too restrictive for empirical data. Measurement error, rounding, missing observations, data corruption, and stochastic processes can produce small violations of the ideal structure.

Define the number of local inversions by

$$I_{\text{local}} = \sum_{i=1}^r \sum_{j=1}^{q-1} \mathbf{1}[x_{ij} > x_{i,j+1}]. \quad (15)$$

The local compliance score is

$$C_{\text{local}} = 1 - \frac{I_{\text{local}}}{r(q-1)}. \quad (16)$$

Similarly, define the number of global score inversions by

$$I_{\text{global}} = \sum_{i=1}^{r-1} \mathbf{1}[S_i < S_{i+1}]. \quad (17)$$

The global compliance score is

$$C_{\text{global}} = 1 - \frac{I_{\text{global}}}{r-1}. \quad (18)$$

A simple composite detection statistic is

$$\boxed{D_q = C_{\text{local}} C_{\text{global}}}. \quad (19)$$

Consequently,

$$0 \leq D_q \leq 1. \quad (20)$$

The value $D_q = 1$ represents exact structural compliance.

5 Weighted Detection

Local and global structure need not have equal importance.

Let

$$0 \leq \lambda \leq 1. \quad (21)$$

A weighted detector may be defined as

$$D_q^{(\lambda)} = \lambda C_{\text{local}} + (1 - \lambda) C_{\text{global}}. \quad (22)$$

Alternatively, a multiplicative detector is

$$D_q^{(\lambda)} = C_{\text{local}}^\lambda C_{\text{global}}^{1-\lambda}. \quad (23)$$

The multiplicative formulation has the useful property that severe failure in either dimension strongly reduces the overall score.

6 Unknown Regiment Size

In realistic data, q is generally unknown.

For an array of length N , the simplest candidate set is

$$\mathcal{Q}(N) = \{q \in \mathbb{N} : q \mid N\}. \quad (24)$$

For each candidate q , compute

$$D_q. \quad (25)$$

The estimated regiment size is

$$\hat{q} = \arg \max_{q \in \mathcal{Q}(N)} D_q. \quad (26)$$

This is a model-selection problem.

If several candidates have similar scores, the data should not be forced into a unique regiment size. Instead, one may report a set of plausible sizes.

7 Detection Architecture

The complete inference procedure can be represented as follows.

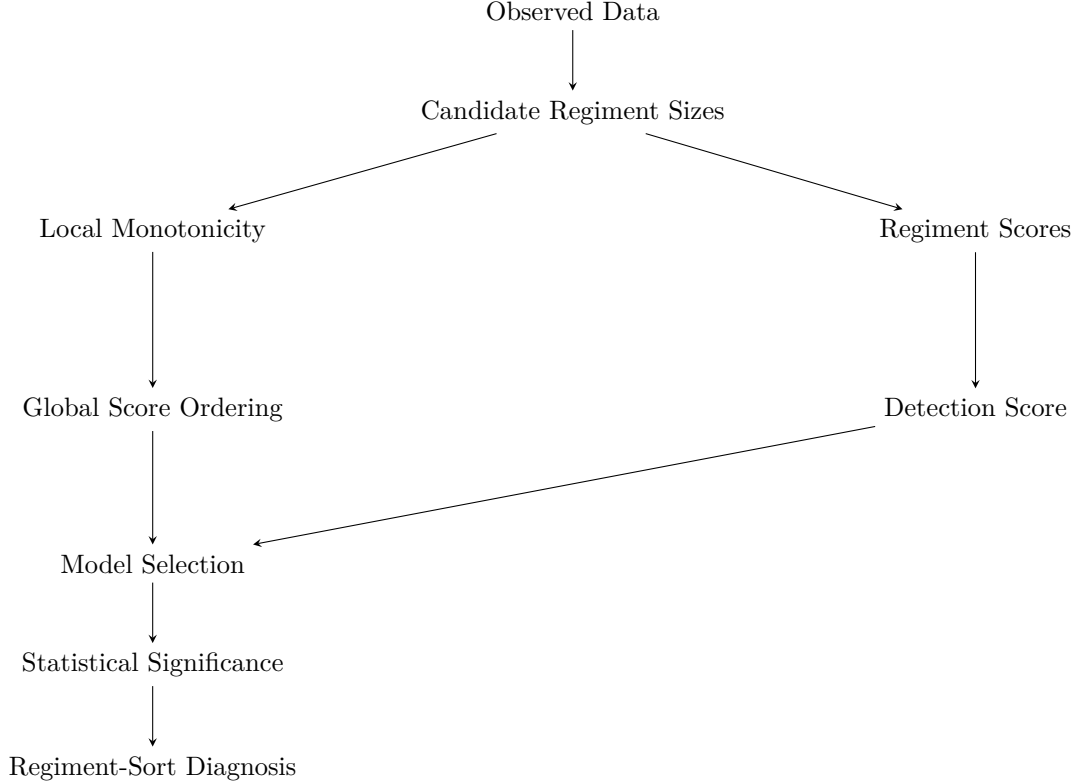


Figure 1: Detection architecture.

The resulting architecture separates descriptive detection from inferential claims.

8 Null Models and Statistical Significance

A high detection score alone does not establish that Regiment Sort caused the observed structure.

Let

$$D_q^{\text{obs}} \quad (27)$$

be the observed detection statistic.

Under a null hypothesis of no Regiment-Sort structure, generate B surrogate datasets,

$$A^{(1)}, \dots, A^{(B)}. \quad (28)$$

Calculate

$$D_q^{(b)} = D_q(A^{(b)}). \quad (29)$$

An empirical upper-tail probability is

$$\hat{p}_q = \frac{1 + \sum_{b=1}^B \mathbf{1}[D_q^{(b)} \geq D_q^{\text{obs}}]}{B+1}. \quad (30)$$

This is a standard permutation-style construction [5, 7].

The null model must preserve whatever features are not under investigation. For independent observations, random permutation may be appropriate. For time-series data, indiscriminate permutation can destroy autocorrelation, seasonality, or volatility clustering. Block permutation, stationary bootstrap, or model-based simulation may therefore be preferable [14].

9 Reconstruction

Detection is followed naturally by reconstruction.

Given $\hat{q}$, define

$$\hat{r} = \frac{N}{\hat{q}}. \quad (31)$$

The estimated latent regiments are

$$\hat{R}_1, \dots, \hat{R}_{\hat{r}}. \quad (32)$$

Their scores are

$$\hat{S}_i = G(\hat{R}_i). \quad (33)$$

The complete inferred structure is therefore

$$\hat{\mathcal{R}} = (\hat{q}, \hat{R}_1, \dots, \hat{R}_{\hat{r}}, \hat{S}_1, \dots, \hat{S}_{\hat{r}}). \quad (34)$$

This is analogous to latent-structure reconstruction in clustering and mixture models [9].

10 Inferring the Aggregation Functional

The sum need not be the actual criterion used to generate an observed pattern.

Consider a family of candidate aggregation functions

$$\mathcal{G} = \{G_1, G_2, \dots, G_K\}. \quad (35)$$

Examples include

$$G_1(R_i) = \sum_j x_{ij}, \quad (36)$$

$$G_2(R_i) = \frac{1}{q} \sum_j x_{ij}, \quad (37)$$

$$G_3(R_i) = \max_j x_{ij}, \quad (38)$$

$$G_4(R_i) = \text{median}(R_i), \quad (39)$$

$$G_5(R_i) = \sum_j w_{ij} x_{ij}. \quad (40)$$

For every (q, G_k) pair, calculate

$$D_{q,k}. \quad (41)$$

The inferred model becomes

$$(\hat{q}, \hat{G}) = \arg \max_{q, G_k} D_{q,k}. \quad (42)$$

This transforms Regiment-Sort detection into a model-selection framework.

Information criteria can also be introduced when the scoring model has different numbers of parameters [1, 16].

11 Robust Detection

Suppose the observed data satisfy

$$x_{ij} = x_{ij}^* + \varepsilon_{ij}, \quad (43)$$

where x_{ij}^* represents latent Regiment-Sort data.

A robust detector should distinguish structural violations caused by measurement noise from genuine structural violations.

One possibility is a tolerance parameter $\delta \geq 0$:

$$I_{\text{local}}(\delta) = \sum_{i,j} \mathbf{1}[x_{ij} > x_{i,j+1} + \delta]. \quad (44)$$

Similarly,

$$I_{\text{global}}(\delta) = \sum_i \mathbf{1}[S_i < S_{i+1} - \delta]. \quad (45)$$

The parameter δ should ideally be linked to measurement uncertainty rather than selected merely to produce a high detection score.

12 Sentinel and Missing Data

The forward Regiment Sort algorithm uses a structural sentinel for padding. Detection must distinguish such structural absence from actual observations.

Let

$$P \quad (46)$$

denote the sentinel.

Define

$$R_i^* = \{x \in R_i : x \neq P\}. \quad (47)$$

The estimated score becomes

$$S_i = G(R_i^*). \quad (48)$$

This distinction is particularly important in empirical datasets, where missing values may be informative rather than random [11].

13 Algorithm

DETECT_REGIMENT_SORT(A)

```

candidate_sizes <- divisors(length(A))

for q in candidate_sizes:
    partition A into regiments of size q

    for each regiment:
        calculate local-order compliance
        calculate aggregate score

    calculate global score compliance

    calculate composite detection score D[q]

q_hat <- candidate q with highest score

```

```

reconstruct candidate regiments

generate null datasets

calculate empirical significance

return:
    q_hat
    regiment structure
    local score
    global score
    detection score
    empirical p-value

```

14 Python Prototype

A minimal implementation is:

```

from typing import Sequence, Dict, Any

def regiment_detection_scores(
    data: Sequence[float],
) -> Dict[int, Dict[str, Any]]:

    data = list(data)
    N = len(data)

    results = {}

    for q in range(1, N + 1):

        if N % q != 0:
            continue

        r = N // q

        regiments = [
            data[i * q:(i + 1) * q]
            for i in range(r)
        ]

        local_violations = 0

        for regiment in regiments:
            local_violations += sum(
                regiment[j] > regiment[j + 1]
                for j in range(q - 1)
            )

        denominator = r * max(q - 1, 1)

        local_score = (
            1.0
            - local_violations / denominator
        )

```

```

    sums = [
        sum(regiment)
        for regiment in regiments
    ]

    global_violations = sum(
        sums[i] < sums[i + 1]
        for i in range(r - 1)
    )

    global_score = (
        1.0
        if r == 1
        else 1.0
        - global_violations / (r - 1)
    )

    detection_score = (
        local_score * global_score
    )

    results[q] = {
        "regiments": regiments,
        "sums": sums,
        "local_score": local_score,
        "global_score": global_score,
        "detection_score": detection_score,
    }

return results

```

The implementation deliberately treats detection as a diagnostic rather than automatically declaring that a dataset was generated by Regiment Sort.

15 Statistical Interpretation

Let

$$H_0 : \text{the observed data contain no Regiment-Sort structure} \quad (49)$$

and

$$H_1 : \text{the observed data contain Regiment-Sort structure.} \quad (50)$$

The detector supplies a test statistic,

$$T(A) = \max_{q \in \mathcal{Q}} D_q(A). \quad (51)$$

The corresponding empirical null distribution is

$$T^{(1)}, \dots, T^{(B)}. \quad (52)$$

The empirical p -value is

$$\hat{p} = \frac{1 + \sum_{b=1}^B \mathbf{1}[T^{(b)} \geq T^{\text{obs}}]}{B + 1}. \quad (53)$$

Thus the problem becomes a formal statistical test rather than a subjective visual inspection.

However, searching over many candidate regiment sizes introduces a multiple-comparison or model-selection issue. The null distribution should therefore replicate the entire search procedure, not merely evaluate the single best candidate after selection.

16 Time-Series Data

For time-series observations,

$$A = (X_1, \dots, X_T), \quad (54)$$

a detected Regiment Sort pattern might represent a recurring temporal aggregation structure. However, serial dependence complicates inference:

$$\text{Cov}(X_t, X_{t+h}) \neq 0. \quad (55)$$

A naive permutation null can therefore produce misleading significance.

Block bootstrap and other dependence-preserving procedures are preferable when temporal structure is substantively important [14].

The same issue occurs in financial returns, macroeconomic panels, sensor data, and physical measurements.

17 Finance and Econometrics

Suppose financial observations are

$$r_t = \frac{P_{t+1}}{P_t} - 1. \quad (56)$$

A sequence of returns can be tested for hierarchical Regiment-Sort structure. For a candidate block size q ,

$$S_i = \sum_{j=1}^q r_{(i-1)q+j}. \quad (57)$$

The score might alternatively be a risk-adjusted quantity,

$$S_i = \frac{\bar{r}_i - r_f}{\sigma_i}. \quad (58)$$

This permits investigation of whether financial data exhibit unusually strong within-block ordering combined with descending block-level performance.

Such a test must be calibrated against appropriate financial null models because financial returns exhibit heteroskedasticity, dependence, and heavy tails [6, 2].

In panel econometrics, a regiment can represent a firm, industry, country, household, or time block. The detection problem then becomes one of identifying latent hierarchical ordering in panel observations [17].

18 Physics and Signal Processing

For a physical signal

$$x(t), \quad (59)$$

a regiment may correspond to a temporal or spatial block. The aggregate score can be energy,

$$E_i = \sum_j x_{ij}^2, \quad (60)$$

rather than amplitude.

Detection then asks whether blocks are internally ordered according to one observable while being globally ordered according to another.

This is related conceptually to segmentation, feature extraction, and multiscale signal analysis.

19 Biology

In biological data, regiments may correspond to cells, species, populations, experimental batches, or temporal cohorts.

Suppose x_{ij} denotes an expression measurement. A group score may be

$$S_i = \sum_j x_{ij}. \quad (61)$$

A Regiment-Sort detector can test whether observations exhibit a combination of local monotonicity and group-level ordering.

However, biological dependence, batch effects, measurement error, and multiple testing must be incorporated into the null model.

20 Chemistry

A regiment may represent a chemical experiment, reaction family, or molecular class.

For reaction yields Y_{ij} ,

$$S_i = \sum_j Y_{ij}. \quad (62)$$

A detected structure could indicate systematic organization in experimental results, but causal interpretation requires an appropriate experimental design.

The distinction between pattern detection and causal inference is essential.

21 Psychology and Psychiatry

Suppose observations represent scores from validated psychological instruments.

A regiment may correspond to a cohort or assessment period, with

$$S_i = \sum_j x_{ij}. \quad (63)$$

Detection can identify organizational structure in the data, but it cannot by itself establish psychological mechanisms or psychiatric diagnoses.

Measurement validity, latent variables, dependence, selection effects, and clinical context must be considered [3, 10].

22 Political Science and International Relations

A state can be represented by a capability vector

$$\mathbf{c}_i = (c_{i1}, \dots, c_{ik}). \quad (64)$$

A scalar score can be constructed as

$$S_i = \mathbf{w}^T \mathbf{c}_i. \quad (65)$$

Detection can then investigate whether states or political units display hierarchical organization consistent with a prescribed ranking mechanism.

This should be understood as statistical description rather than an assumption that political power is reducible to a single scalar [15, 12].

23 Welfare Economics

Let x_{ij} represent welfare-relevant resources.

An aggregate resource score is

$$S_i = \sum_j x_{ij}. \quad (66)$$

A welfare score may instead be

$$W_i = \sum_j u(x_{ij}), \quad (67)$$

with

$$u'(x) > 0, \quad u''(x) < 0. \quad (68)$$

The inferred regiment structure may therefore depend on whether the researcher seeks to detect resource aggregation or welfare aggregation.

This illustrates a broader principle: detection is conditional upon the measurement and aggregation model.

24 Warfare Economics and Abstract Force Analysis

The original terminology naturally admits an abstract force-allocation interpretation.

Let x_{ij} represent an abstract capability measure and let

$$S_i = \sum_j x_{ij}. \quad (69)$$

The detector asks whether observed formations exhibit internal ordering and descending aggregate capability.

Real force effectiveness, however, is multidimensional. A more general model is

$$S_i = F(L_i, R_i, T_i, I_i, C_i, M_i), \quad (70)$$

where the variables represent logistics, readiness, terrain, information, coordination, and mission compatibility.

Consequently, detected Regiment-Sort structure should be interpreted as a property of the chosen measurement system, not as a complete model of military effectiveness.

25 AI and Machine Learning

The detection problem can itself become a machine-learning problem.

Let

$$\phi(A; q) = (C_{\text{local}}, C_{\text{global}}, D_q) \quad (71)$$

be a feature representation.

A classifier can estimate

$$\Pr(H_1 \mid \phi(A; q)). \quad (72)$$

Synthetic Regiment-Sort datasets can be generated for supervised learning, while real datasets can provide out-of-distribution testing.

Alternatively, an autoencoder or representation-learning system could learn features associated with hierarchical ordering [8].

Machine learning does not remove the need for statistical identification: a classifier can detect predictive structure without establishing that the structure is causal.

Table 1: Model diagnostics

Diagnostic	Question	Interpretation
Local score	Are regiments internally ascending?	Local structure
Global score	Are regiment scores descending?	Group structure
Composite score	Do both hold simultaneously?	Overall signature
Candidate size	Which q fits best?	Structural identification
Null probability	Is the score unusual?	Statistical evidence
Robust score	Does the pattern survive noise?	Structural stability
Aggregation test	Which G works best?	Criterion identification

26 Causal Inference

A crucial distinction is

$$\text{Regiment-Sort structure} \neq \text{Regiment-Sort causation}. \quad (73)$$

If an observed dataset exhibits the pattern, this does not establish that a Regiment Sort procedure generated it.

A causal claim requires additional information concerning the data-generating process, interventions, or experimental design [13].

The appropriate hierarchy is therefore

$$\text{detection} \rightarrow \text{identification} \rightarrow \text{explanation} \rightarrow \text{causal inference}. \quad (74)$$

Each step requires stronger assumptions than the preceding step.

27 Model Diagnostics

A useful empirical diagnostic table is:

28 Theoretical Properties

Proposition 1. If an array is exactly produced by Regiment Sort using regiment size q , then

$$D_q^{\text{exact}} = 1. \quad (75)$$

Proof. By construction, every regiment is internally ascending and the regiment scores are descending. Hence every local indicator and the global indicator equals one. $\square$

Proposition 2. For every dataset and candidate q ,

$$0 \leq D_q \leq 1. \quad (76)$$

Proof. Both C_{local} and C_{global} are normalized fractions lying in $[0, 1]$. Their product therefore lies in $[0, 1]$. $\square$

Proposition 3. A high detection score does not logically imply that Regiment Sort generated the data.

Proof. The detector is a function of observable ordering properties. Distinct data-generating processes can generate identical ordering properties. Therefore structural equivalence does not imply causal equivalence. $\square$

29 A General Statistical Framework

The complete detection framework can be represented by

$$\mathcal{D} = (A, \mathcal{Q}, \mathcal{G}, \mathcal{H}), \quad (77)$$

where $\mathcal{Q}$ is the candidate regiment-size space, $\mathcal{G}$ is the candidate aggregation-function space, and $\mathcal{H}$ specifies the null and error models.

The optimal detected structure is

$$(\hat{q}, \hat{G}) = \arg \max_{q, G} D(A; q, G). \quad (78)$$

Statistical significance is then evaluated by repeating the entire selection process under the null:

$$T^{(b)} = \max_{q, G} D(A^{(b)}; q, G). \quad (79)$$

This is important because the researcher is searching over models rather than testing one model selected in advance.

30 Discussion

The forward and inverse problems can now be written together:

$$\text{Forward: } A \longrightarrow \text{RS}(A), \quad (80)$$

$$\text{Inverse: } A \longrightarrow (\hat{q}, \hat{G}, \hat{\mathcal{R}}, \hat{p}). \quad (81)$$

The first is an algorithmic transformation. The second is a statistical inference problem. This distinction substantially expands the scope of Regiment Sort. It allows researchers to ask not merely

$$\text{“How do we perform Regiment Sort?”} \quad (82)$$

but

$$\boxed{\text{“Does the structure of this dataset look as though Regiment Sort could have produced it?”}} \quad (83)$$

The latter question is potentially useful wherever observations possess hierarchical structure.

31 Conclusion

Regiment Sort detection is an inverse hierarchical-ordering problem.

Its fundamental structure is

$$\boxed{\begin{array}{ccccccc} \textit{observed} & \rightarrow & \textit{candidate} & \rightarrow & \textit{local} & \rightarrow & \textit{aggregate} & \rightarrow & \textit{global} & \rightarrow & \textit{statistical} \\ \textit{data} & & \textit{groups} & & \textit{ordering} & & \textit{scores} & & \textit{ordering} & & \textit{test} \end{array}}. \quad (84)$$

The framework provides five principal inferential tasks:

1. detect Regiment-Sort structure;
2. identify the most plausible regiment size;
3. reconstruct the latent regiments;
4. infer the aggregation criterion;
5. evaluate statistical significance and robustness.

The central methodological lesson is that a high structural score should be regarded as evidence of an ordering pattern, not automatically as evidence of a particular causal mechanism.

Regiment Sort thus becomes more than a sorting algorithm. It becomes a candidate statistical structure that can be searched for, measured, compared, tested, and reconstructed across heterogeneous scientific and economic datasets.

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Glossary

Aggregation Function

A function mapping observations within a regiment to a group-level score.

Candidate Regiment Size

A proposed number of observations contained in each regiment.

Composite Detection Score

A statistic combining local and global evidence for Regiment-Sort structure.

Global Compliance

The degree to which regiment scores decrease from one regiment to the next.

Local Compliance

The degree to which observations within each regiment are ascending.

Null Model

A probabilistic model representing the absence of the hypothesized Regiment-Sort structure.

Permutation Test

A statistical procedure based on repeatedly rearranging or resampling data under a specified null hypothesis.

Reconstruction

Recovery of candidate regiment boundaries and scores from observed data.

Regiment

A fixed-size group of observations under the Regiment-Sort framework.

Regiment-Sort Detection

The statistical assessment of whether observed data exhibit the defining hierarchical ordering properties of Regiment Sort.

Robust Detection

Detection that explicitly allows for measurement noise or bounded violations of ideal ordering.

Sentinel

A structural marker representing an absent or padded position.

Stable Ordering

An ordering rule that preserves the original relative order of equal-score regiments.

Structural Signature

A measurable collection of properties characteristic of a hypothesized data-generating or data-transforming procedure.

Structural Inference

Inference concerning hidden organization or generating structure from observable data.

The End